Practice workbook · Linear inequalities and the 'both' overlap

Solving a single inequality in which x appears only to the first power, including one with surd coefficients.

15 question parts: 1 worked-example retry and 14 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-IDN-01

1. Question 1 · part a · 2 marks

2023 JAN · Paper 02

Worked-example retry · Attempt part a only.

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1 Find the set of values of x for which

(a) $2(3x - 1) < 4 - 3x$

(2)

(b) $3x^2 - 8x - 3 < 0$

(4)

(c) **both** $2(3x - 1) < 4 - 3x$ **and** $3x^2 - 8x - 3 < 0$

(1)

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(Total for Question 1 is 7 marks)

1. Question 1 · part a · 2 marks · Working and answer

Working and answer

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2. Question 1 · part a · 1 mark

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

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1 Find the set of values for x for which

(a) $5x - 10 > 4x - 7$

(1)

(b) $2x^2 - 11x + 5 < 0$

(3)

(c) **both** $5x - 10 > 4x - 7$ **and** $2x^2 - 11x + 5 < 0$

(1)

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(Total for Question 1 is 5 marks)

2. Question 1 · part a · 1 mark · Working and answer

Working and answer

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3. Question 2 · part a · 1 mark

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Further practice · Attempt part a only.

2 Find the set of values of x for which

(a) $3x - 8 < 5x + 3$

(1)

(b) $4x^2 - 7x + 1 > 6 - 2x^2$

(4)

(c) **both** $3x - 8 < 5x + 3$ **and** $4x^2 - 7x + 1 > 6 - 2x^2$

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3. Question 2 · part a · 1 mark · Working and answer

Working and answer

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4. Question 2 · part a · 1 mark

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Further practice · Attempt part a only.

2 Find the set of values for x for which

(a) $9-3x>11x+2$

(1)

(b) $10x^2+7x<12$

(3)

(c) **both** $9-3x>11x+2$ **and** $10x^2+7x<12$

(1)

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4. Question 2 · part a · 1 mark · Working and answer

Working and answer

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5. Question 1 · part a · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part a only.

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1 Find the set of values for x for which

(a) $8x - 7 < 5x + 5$

(2)

(b) $2x^2 - 5x - 3 > 0$

(3)

(c) **both** $8x - 7 < 5x + 5$ **and** $2x^2 - 5x - 3 > 0$

(1)

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(Total for Question 1 is 6 marks)

5. Question 1 · part a · 2 marks · Working and answer

Working and answer

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6. Question 2 · part a · 2 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

2 Find the set of values of x for which

(a) $2(x+1) < 5x - 2$

(2)

(b) $3x^2 - x \leqslant 10$

(3)

(c) **both** $2(x+1) < 5x - 2$ **and** $3x^2 - x \leqslant 10$

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6. Question 2 · part a · 2 marks · Working and answer

Working and answer

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7. Question 2 · part a · 1 mark

2020 JAN · Paper 02

Further practice · Attempt part a only.

2 Find the set of values of x for which

(a) $3 + 2x \leqslant x + 2$

(1)

(b) $8x^2 + 10x < 3$

(4)

(c) **both** $3 + 2x \leqslant x + 2$ **and** $8x^2 + 10x < 3$

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7. Question 2 · part a · 1 mark · Working and answer

Working and answer

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8. Question 1 · part whole · 4 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part whole only.

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1 Without using a calculator, solve the inequality $\sqrt{50}\,x - \sqrt{18} > 6x + 5$

Give your answer in an exact form with a rationalised denominator.
Show your working clearly.

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(Total for Question 1 is 4 marks)

8. Question 1 · part whole · 4 marks · Working and answer

Working and answer

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9. Question 1 · part c · 1 mark

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

1 Find the set of values for x for which

(a) $5x - 10 > 4x - 7$

(1)

(b) $2x^2 - 11x + 5 < 0$

(3)

(c) **both** $5x - 10 > 4x - 7$ **and** $2x^2 - 11x + 5 < 0$

(1)

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(Total for Question 1 is 5 marks)

9. Question 1 · part c · 1 mark · Working and answer

Working and answer

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10. Question 2 · part c · 1 mark

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

2 Find the set of values of x for which

(a) $3x - 8 < 5x + 3$

(1)

(b) $4x^2 - 7x + 1 > 6 - 2x^2$

(4)

(c) **both** $3x - 8 < 5x + 3$ **and** $4x^2 - 7x + 1 > 6 - 2x^2$

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10. Question 2 · part c · 1 mark

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Further practice · Attempt part c only.

Question 2 continued

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(Total for Question 2 is 6 marks)

10. Question 2 · part c · 1 mark · Working and answer

Working and answer

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11. Question 1 · part c · 1 mark

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Further practice · Attempt part c only.

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(Total for Question 1 is 7 marks)

11. Question 1 · part c · 1 mark · Working and answer

Working and answer

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12. Question 2 · part c · 1 mark

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Further practice · Attempt part c only.

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(1)

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12. Question 2 · part c · 1 mark

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Question 2 continued

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(Total for Question 2 is 5 marks)

12. Question 2 · part c · 1 mark · Working and answer

Working and answer

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13. Question 1 · part c · 1 mark

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Further practice · Attempt part c only.

1 Find the set of values for x for which

(a) $8x - 7 < 5x + 5$

(2)

(b) $2x^2 - 5x - 3 > 0$

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(Total for Question 1 is 6 marks)

13. Question 1 · part c · 1 mark · Working and answer

Working and answer

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14. Question 2 · part c · 1 mark

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Further practice · Attempt part c only.

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(a) $2(x+1) < 5x - 2$

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14. Question 2 · part c · 1 mark

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Question 2 continued

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14. Question 2 · part c · 1 mark · Working and answer

Working and answer

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15. Question 2 · part c · 1 mark

2020 JAN · Paper 02

Further practice · Attempt part c only.

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(1)

(b) $8x^2 + 10x < 3$

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15. Question 2 · part c · 1 mark

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15. Question 2 · part c · 1 mark · Working and answer

Working and answer

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